MEET BRICKBUDDY

Show the pile.
Pick a build. Change anything.

An AI building companion for children aged 6–8. Scan your bricks, pick a robot, then change it by voice or tap — and watch the instruction manual rewrite itself in real time.

THE BOOKLET

One-way paper.

- ✕ Can't see what bricks the kid actually has
- ✕ Can't answer "why?" or adapt mid-build
- ✕ Can't recolor when they want something different
- ✕ Only shows one "correct" model

→

BRICKBUDDY

A living manual.

- ✓ Scans the pile, recommends a build
- ✓ "Make it blue" — 3D and manual rewrite themselves
- ✓ Tap a brick to recolor only that one
- ✓ Teaches STEAM through every step

EVERY SESSION FOLLOWS ONE ARC · FIVE SCREENS, ZERO MENUS

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STAGE 1

Discover

Camera-first. Scan the pile, tell Buddy out loud, or just browse. One primary action, no text fields.

CAMERA · VOICE

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STAGE 2

Inventory

An animated scan reveals every brick by color and count, then ranks three matched builds on what the kid actually owns.

SCAN · RECOMMEND

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STAGE 3

Build ★

3D viewer + paper manual. Say *"make it blue"* or tap a brick — both 3D and manual rewrite themselves in real time.

3D · VOICE · TAP

🔬

STAGE 4

Learn

Five STEAM tabs grounded in what was just built. A two-question quiz lights up achievements, not grades.

SCI · TECH · ENG · ART · MATH

🏆

STAGE 5

Celebrate

Confetti, a rotating 3D trophy, a build report, and a printable certificate with the child's name on it.

REPORT · CERTIFICATE

WHAT MAKES IT WORK · THREE HONEST CLAIMS

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Model and manual share one source

Voice, tap, and structural edits flow through a single mutation engine. The 3D view, the printed manual, and the change log all read from the same data — never out of sync.

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Every brick is data

Models are stored as structured brick data. Inventory counts, manual steps, and 3D rendering all read from the same source — adding a model is one JSON file, not rewriting screens.

🪶

Runs from a URL

Single-page web app on Cloudflare's edge. No app store, no install, no parent setup. Open the link, point a webcam, build something. Works on a Chromebook in a school.

Y

Yusong Huang

FULL IMPLEMENTATION · MATERIALS

Engineered every screen — React, 3D viewer, AI pipeline, Cloudflare deployment — and produced this demo + poster + report.

J

Jiayi Huang

IDEATION · MATERIALS

Brought the original BrickBuddy concept, user research with children 6–8, and co-produced the demo materials.

LIVE `brickbuddy.pages.dev`

REPO `github.com/Yusong-Enceladus/CMAA5043`

DEMO VIDEO `demo/video/BrickBuddy_investor_demo.mp4`

BUILT WITH React 19 · R3F · Three.js · Cloudflare Pages

SCAN → TRY IT

BRICKBUDDY · AI BUILDING COMPANION · AGES 6–8

Yusong Huang · Jiayi Huang